

Chess Insights: Result Predictors and Play Sequences by Player Ranking

Description

This project will explore online chess match data from Lichess.org. The main goal is to identify which game characteristics, opening moves, and move sequences are associated with wins and losses. This project will also examine whether result predictors and common move sequences vary across different player rating levels. Sample questions include:

- What are the strongest individual predictors of wins and losses?
- What are strong aggregate predictors of wins and losses?
- What are the most common opening moves, and what is the win/loss breakdown for those moves?
- Is it possible to identify sequences of k-moves that are strongly associated with game outcomes?
- Do the answers to the questions above vary by player ranking?

Prior Work

There are a number of related projects listed on the Lichess.org website. I explored several and have summarized three that appear most related to the direction of my research below:

- Chess blunders examines whether the Lichess low clock alarm causes more blunders. (<https://github.com/Antiochian/chess-blunders/blob/main/README.md>)
- Chessroots.com used Lichess data as well as others to examine opening moves and following sequences. The website provides a visual tool to move through move sequences and display the frequency of that path, as well as the win/loss percentage associated. (<https://www.chessroots.com/>)
- Chess Opening Stats analyzes opening moves and associated win rates. (<https://github.com/Paul566/chessOpeningStats>)

Dataset

The dataset is the Lichess dataset from February 2026. I downloaded it and confirmed that my proposed questions can be answered with the data. The file was 27.8GB compressed, and 184.12GB decompressed. It contains information for 84,600,043 chess games. The dataset citation is provided in MLA format below:

Lichess. "Lichess Database: Standard Rated Games, February 2026." Lichess, 2026, <https://database.lichess.org/>.

Proposed Work

Data Cleaning- The data will be parsed from the plain text format into a structured format suitable for analysis such as a parquet file. I will check for and remove games that are empty or incomplete, games that have invalid ratings or outcomes, and games that are unable to be parsed correctly. I will ensure that the contents of each attribute are uniform and standardized.

Data Preprocessing- The tags in the plain text format translate to attributes for each game. There is also opportunity to engineer additional features such as number of moves, time breakdown per

player, and player rating bin. It might be necessary to create a new table specifically for move sequence analysis. During pre-processing I will also create a small subset dataset for testing code and validating analysis steps before running the entire dataset.

Data integration- I do not plan to integrate additional external data to this dataset. The dataset already contains the game metadata, player data, results, and move sequences necessary for analysis.

Analysis and Modeling- Analysis will include descriptive statistics, grouped comparisons, and cluster analysis. Descriptive statistics will be used to examine opening moves, move sequence, and differences between player rating bins. Predictive modeling will be used to test whether individual or aggregate features can predict game outcomes.

Visualization- Interesting results will be supported by visualizations such as graphs and tables where appropriate.

Tools

I plan to primarily use Python and associated libraries for the entire project. I have not explored very deeply into some of the data analysis or machine learning libraries, so this project provides a good opportunity to do so. I will use some familiar libraries such as pandas, NumPy, matplotlib, and seaborn for basic data operations and visualization. I will use parquet files to store the dataset. I will most likely use the libraries below as well:

- python-chess for parsing PGN files and extracting moves, openings, and game metadata
- scikit-learn for predictive modeling and evaluation
- zstandard for working with compressed .zst files if needed

Evaluation

At each stage in the project, I plan to evaluate my use of statistical rigor and best practice data analysis methodologies. Areas of uncertainty or possible weakness in the data or results will be documented and addressed when possible. For descriptive analysis, I will ensure sample sizes are large enough to support meaningful conclusions. During predictive modeling, best practices such as train/test splits and evaluation of results using standard metrics (accuracy, precision, recall, F1 score, confusion matrices, etc) will be enforced.

The overall project will be evaluated with two criteria. First, I will evaluate the project's results first in terms of answering the proposed questions and whether I was able to identify and present any interesting findings. Second, I will evaluate the project in terms of learning value and skills gained.